Dirac-to-Tensor-Curl Eta Transfer on Round Spherical Space Forms with Affine-ADE Specializations

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1 October 2026

Abstract

Let $G < SO(4)$ be finite and effective, act freely on the unit round three-sphere, and fix a lift of G to $\mathrm{Spin}(4) = SU(2)_L \times SU(2)_R$. On the resulting spherical space form, for every integer $J \geq 1$ and every flat bundle associated to an irreducible representation of G, we consider the eta invariant of the helicity-signed transverse-traceless tensor-curl spectral family of rank J. If $\boldsymbol{\eta}_C^D$ is the twisted spin-Dirac eta vector and A_m^L, A_m^R denote multiplication in the representation ring of G by the restricted symmetric powers of the two lifted defining representations, then

$$\boldsymbol{\eta}_J^{\mathrm{TT}} = -\frac{1}{2}\left(A_{2J-1}^L + A_{2J-1}^R\right)\boldsymbol{\eta}_C^D.$$

The proof combines a two-variable character identity with a continuation lemma for finite-order Hurwitz–Lerch series and requires neither commutativity of G nor a one-sided action. For one-sided subgroups of $SU(2)$, the transfer specializes along the McKay correspondence: the eta vectors satisfy a rank recurrence governed by the affine-ADE adjacency matrix with an explicit linear-plus-periodic source. We give closed formulas for the cyclic and binary-dihedral families and specify the recurrence and source periods for the exceptional binary polyhedral families. The two-sided cyclic case covers every lens space $L(p, q)$, while the nonabelian direct product $Q_8 \times \mathbb{Z}_3$ illustrates the general statement. At rank one the identity is consistent with the classical modulo-integers relation between curl and Dirac eta invariants, with the Dirac sign specified below. The contribution here is the exact all-rank, flat-twisted transfer and its McKay-level organization.

1 Introduction

Eta invariants measure the signed imbalance of a self-adjoint spectrum. Their classical uses include index theory on manifolds with boundary, rho-invariants, and phases of spectral determinants [APS75, Don78, Gil84]. On spherical space forms, twisted spin-Dirac eta invariants are well developed: character formulas, Molien-series encodings, and complete spin-$\frac{1}{2}$ tables for the finite subgroups of $SU(2)$ are already in the literature [CM01, Deg09, Deg01]. The representation theory behind the affine-ADE diagrams is likewise classical [McK80, Kos85, Ros03].

The operator studied here is different. On the round sphere we use the helicity-signed, projected transverse-traceless branch of the higher-spin self-gradient family described by Homma and Homma–Tomihisa [Hom01, HT21]. We denote its convention-fixed spectral realization by $\mathcal{C}_J$. At $J = 1$ it is the classical coexact curl operator; for $J \geq 2$ the argument below uses the round-sphere

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module decomposition and signed eigenvalues recorded in (3)–(4). The rank-two massive-gravity literature records the corresponding unsigned spectrum while leaving its phase choice unresolved [Hom01, HT21, CLM11, BGVZ11]. We retain the helicity sign, analytically continue the imbalance, and organize all flat twists by the representation ring of the deck group.

The rank-one member is the classical coexact curl sector, whose spectral and eta-theoretic setting is already established [APS75, Bär19]. The contribution begins at $J \geq 2$ with the all-rank transfer and its McKay organization; we do not claim novelty for curl itself.

The main result is not merely a list of exceptional-group values. It is a universal transfer from a classical spin-Dirac eta vector to every integer-rank projected-TT eta vector on an oriented round spherical space form with a chosen Spin(4) lift. The proof is classwise and does not assume that the deck group is cyclic or one-sided. For one-sided finite subgroups of $SU(2)$, this transfer turns the McKay graph into an eta-level computational object and yields a sourced rank recurrence. The cyclic, binary-dihedral, and exceptional formulas then become explicit specializations of the general identity.

The theorem concerns this round-sphere spectral family and its eta invariant; it is not an operator intertwining or isospectrality statement. The affine-ADE labels organize the one-sided representation theory, without any physical interpretation or selection of a group, representation, or topology.

This paper is part of a series. The lens-space conventions and closed-form evaluations to which the cyclic corollary applies are developed in [Mah26b]; the bivariate Chebyshev–McKay algebra underlying the recurrences is treated in greater generality in [Mah26a]; and the realization of $\mathcal{C}_J$ within a four-dimensional boundary problem, together with the resulting APS corollaries, appears in [Mah26c]. The present proof is self-contained and uses none of those companion results.

Section 2 fixes the operator and continuation conventions. Section 3 proves the general transfer and its one-sided specialization. The remaining sections treat lens spaces, a mixed nonabelian example, and the affine-ADE recurrences before discussing scope and open problems.

2 Operator, quotient, and twisting conventions

Identify S^3 with $SU(2)$. Let $G < SO(4)$ be finite and effective, choose a spin lift

$$\tau = (\tau_L, \tau_R) : G \longrightarrow SU(2)_L \times SU(2)_R, \qquad g \cdot x = \tau_L(g) x \tau_R(g)^{-1}, \tag{1}$$

and assume the action is free. Equivalently, for every $g \neq e$,

$$\chi_1(\tau_L(g)) \neq \chi_1(\tau_R(g)). \tag{2}$$

The ineffective central pair $(-1,-1)$ is therefore absent from $\tau(G)$. Put $M_G = S^3/G$, with the unit round metric and orientation inherited from the boundary of the standard oriented ball in $\mathbb{C}^2$. The positive-left-heavy convention below agrees with [Mah26b, Mah26c]; in the latter's full boundary operator the auxiliary finite character is subtracted.

Proposition 2.1 (Lifts and spin structures)**.** *Let $\pi : \mathrm{Spin}(4) \to SO(4)$ be the double covering. A lift in (1) is a homomorphic section of the pulled-back central extension*

$$1 \longrightarrow \{(1,1),(-1,-1)\} \longrightarrow \pi^{-1}(G) \longrightarrow G \longrightarrow 1.$$

For a free effective action, such lifts are equivalent to spin structures on M_G. They exist because every oriented three-manifold is spin, and the set of lifts is a torsor for

$$H^1(M_G; \mathbb{Z}_2) \cong \mathrm{Hom}(G, \mathbb{Z}_2).$$

Every lift is automatically injective. If instead one starts with a subgroup of $\mathrm{Spin}(4)$ *containing* $(-1,-1)$*, its action on* S^3 *is ineffective and the effective deck group is the quotient by that kernel.*

Proof. A spin structure on the quotient pulls back to the unique spin structure on S^3 and lifts the deck transformations compatibly with composition; conversely, a homomorphic lift descends the spin bundle. The ratio of two lifts takes values in the central kernel and is a homomorphism $G \to \mathbb{Z}_2$, while multiplying a lift by any such character produces another lift. Since $\pi \circ \tau$ is the inclusion of the effective group, $\tau(g) = 1$ implies $g = 1$. The last assertion is the kernel statement for the Spin(4) action. See also the spherical-space-form spin conventions in [Bär92, Bär96]. □

Write $V_n = \mathrm{Sym}^n \mathbb{C}^2$ for the irreducible $SU(2)$ representation of highest weight n and dimension $n+1$. For integer $J \geq 1$, put

$$\mathcal{H}_J = \widehat{\bigoplus_{n\geq 0}} \left(F^+_{J,n} \oplus F^-_{J,n} \right), \qquad F^+_{J,n} = V_{n+2J} \boxtimes V_n, \quad F^-_{J,n} = V_n \boxtimes V_{n+2J}. \tag{3}$$

The completed sum uses the invariant Hilbert norms on its finite-dimensional summands. Define the signed spectral operator $\mathcal{C}_J$ by

$$\begin{aligned} \mathcal{C}_J|_{F^+_{J,n}} &= +(n+J+1), \\ \mathcal{C}_J|_{F^-_{J,n}} &= -(n+J+1), \qquad n \geq 0. \end{aligned} \tag{4}$$

Equations (3)–(4) define the representation-theoretic spectral realization used below. The cited higher-spin sources supply the round-sphere module decomposition and eigenvalue normalization [Hom01, HT21]. For $J = 1$ this is the familiar coexact-curl branch. For higher rank we refer to it as the projected TT tensor-curl family in this convention-fixed sense; no extension to a different metric or to an unprojected tensor operator is asserted.

Proposition 2.2 (Analytic foundation)**.** *The operator* $\mathcal{C}_J$ *with domain*

$$\mathrm{Dom}(\mathcal{C}_J) = \left\{ v = (v^+_n, v^-_n) : \sum_{n\geq 0} (n+J+1)^2 (\|v^+_n\|^2 + \|v^-_n\|^2) < \infty \right\}$$

is self-adjoint, has compact resolvent, and has no zero modes. The left and right $SU(2)$ *actions preserve its domain and commute with it. Hence the induced operator on the multiplicity space*

$$\mathcal{H}_{J,\rho} = \mathrm{Hom}_G(V_\rho, \mathcal{H}_J) \cong (\mathcal{H}_J \otimes V^*_\rho)^G \tag{5}$$

is self-adjoint with compact resolvent for every quotient satisfying (1)–(2)*.*

Proof. The orthogonal summands in (3) are finite-dimensional, the displayed eigenvalues are real, and their absolute values tend to infinity. The stated weighted-square-summability domain is therefore the maximal domain of the diagonal self-adjoint operator, and its resolvent is compact. Equivariance is immediate because each summand is an $SU(2) \times SU(2)$ module on which $\mathcal{C}_J$ is scalar. Finite-group isotypic projections commute with the operator and its resolvent. Passing to the displayed multiplicity space, rather than the full isotypic summand $V_\rho \otimes \mathrm{Hom}_G(V_\rho, \mathcal{H}_J)$, is exactly the copy-multiplicity normalization used below and introduces no extra factor of $\dim \rho$. □

For $\rho \in \mathrm{Irr}(G)$, the associated-bundle normalization is the copy multiplicity

$$m_\rho(W) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_\rho(g)} \chi_W(g). \tag{6}$$

There is no additional factor of $\dim \rho$. Replacing the associated-bundle convention by its dual relabels ρ by ρ^* and changes no vector identity when done consistently.

The full twisted eta invariant is

$$\eta^{\mathrm{TT}}_{J,\rho}(s) = \sum_{n \geq 0} \frac{m_\rho(F^+_{J,n}) - m_\rho(F^-_{J,n})}{(n+J+1)^s}, \qquad \eta^{\mathrm{TT}}_{J,\rho} = \eta^{\mathrm{TT}}_{J,\rho}(0). \tag{7}$$

The equality at $s = 0$ means meromorphic continuation. The operator has no zero modes in these conventions.

The proof architecture is deliberately separated into four steps:

(i) express each nonidentity class contribution as a finite root-of-unity exponential–polynomial sum;

(ii) continue it by residue classes, proving that its value at $s = 0$ is the Abel value and is independent of the positive spectral shift;

(iii) compare the Dirac and TT rational generating functions at $t = 1$; and

(iv) project the resulting class-function identity into irreducible copy-multiplicity coordinates.

This shift-independence step is what permits the Dirac origin $n + \frac{3}{2}$ and the TT origin $n + J + 1$ to enter one eta identity.

Lemma 2.3 (Finite-order Hurwitz–Lerch continuation)**.** *For a nonidentity class of a free effective finite action, each signed Dirac or TT shell coefficient is a finite sum*

$$\sum_{\lambda} \sum_{k=0}^{2} c_{\lambda,k} n^k \lambda^n, \qquad \lambda^{|G|} = 1, \quad \lambda \neq 1. \tag{8}$$

Its eta continuation is a finite sum of shifted Hurwitz–Lerch functions $\Phi(\lambda, s-j, a)$ and is regular at $s = 0$. Moreover,

$$\Phi(\lambda, 0, a) = \frac{1}{1-\lambda}, \qquad \lambda \neq 1, \tag{9}$$

independently of the positive spectral shift a. Consequently the eta value at zero equals the Abel value of the corresponding rational generating function at $t = 1$.

Proof. The two eigenvalues of each lifted factor are roots of unity. A noncentral $SU(2)$ character is a sum of two exponentials in n, while a Weyl limit at ± 1 contributes at most one factor linear in n. Products of the two factor characters therefore have the form (8) with degree at most two. Freeness gives $z \neq w$ and $zw \neq 1$ in eigenvalue coordinates for every nonidentity class; equivalently, none of the four phases $zw, z/w, w/z, (zw)^{-1}$ that survives the signed difference equals one. If exactly one of z, w is central, the same conclusion follows by the corresponding single Weyl limit; freeness keeps the remaining denominator nonzero. The mixed fully central cases have phase -1. The identity shells cancel termwise, and the ineffective pair $(-1, -1)$ is excluded.

If q is the order of λ, decomposition into residue classes gives

$$\Phi(\lambda, s, a) = q^{-s} \sum_{r=0}^{q-1} \lambda^r \zeta\left(s, \frac{a+r}{q}\right). \tag{10}$$

This formula, not the defining power series, is the meaning of $\Phi(\lambda, s, a)$ at a root of unity on $|\lambda| = 1$. The possible pole at $s = 1$ cancels because $\sum_{r=0}^{q-1} \lambda^r = 0$. Using $\zeta(0, x) = \frac{1}{2} - x$ and $\sum_{r=0}^{q-1} r\lambda^r = -q/(1-\lambda)$ gives (9). For the polynomial weights use the finite identity

$$\sum_{n\geq 0} n^k \lambda^n (n+a)^{-s} = \sum_{j=0}^{k} \binom{k}{j} (-a)^{k-j} \Phi(\lambda, s-j, a),$$

initially where the series converges and then by continuation. No differentiation with respect to a root of unity is needed. At nonpositive integers the Bernoulli-polynomial formula for Hurwitz zeta gives the Abel value of each polynomial-geometric series. In particular, the value at $s = 0$ is the Abel value of $\sum n^k \lambda^n$, independent of a. This proves regularity at zero and independence from the Dirac or TT shift. □

For the spin-Dirac control, use the branches

$$D_n^+ = V_n \boxtimes V_{n+1}, \qquad D_n^- = V_{n+1} \boxtimes V_n, \tag{11}$$

with eigenvalues $\pm(n + 3/2)$, and put

$$\eta_{C,\rho}^D(s) = \sum_{n\geq 0} \frac{m_\rho(D_n^+) - m_\rho(D_n^-)}{(n+3/2)^s}. \tag{12}$$

On a one-sided quotient this is the sign printed in Cisneros–Molina's spectral convention: the positive branch carries V_n in $n + 2$ copies and the negative branch carries V_{n+1} in $n + 1$ copies [CM01]. We denote the resulting full twisted eta vector by $\boldsymbol{\eta}_C^D$. This fixes the Dirac sign convention used throughout.

3 General Spin(4) Dirac-to-tensor-curl transfer

For $g \in G$, write z, z^{-1} and w, w^{-1} for the defining eigenvalues of $\tau_L(g)$ and $\tau_R(g)$. For $\Re s > 3$, define the equivariant Dirichlet series

$$d_C(g; s) = \sum_{n\geq 0} \frac{\chi_n(z)\chi_{n+1}(w) - \chi_{n+1}(z)\chi_n(w)}{(n+3/2)^s}, \tag{13}$$

$$t_J(g; s) = \sum_{n\geq 0} \frac{\chi_{n+2J}(z)\chi_n(w) - \chi_n(z)\chi_{n+2J}(w)}{(n+J+1)^s}. \tag{14}$$

Put $d_C(g) = d_C(g; 0)$ and $\epsilon_J(g) = t_J(g; 0)$ by the continuation of Lemma 2.3.

Lemma 3.1 (Two-variable class identity)**.** *For noncentral eigenvalue coordinates, set*

$$P_{a,b}(z, w; t) = \sum_{n\geq 0} \chi_{n+a}(z)\chi_{n+b}(w)t^n. \tag{15}$$

For $|t| < 1$,

$$P_{a,b}(z,w;t) = \frac{1}{(z-z^{-1})(w-w^{-1})}\Bigg[\frac{z^{a+1}w^{b+1}}{1-tzw} - \frac{z^{a+1}w^{-b-1}}{1-tz/w} - \frac{z^{-a-1}w^{b+1}}{1-tw/z} + \frac{z^{-a-1}w^{-b-1}}{1-t/(zw)}\Bigg]. \tag{16}$$

At $t = 1$, freeness makes the simplified denominators nonzero and

$$P_{0,1} - P_{1,0} = -\frac{2zw}{(z-w)(zw-1)} = \frac{2}{\chi_1(w) - \chi_1(z)}, \tag{17}$$

$$P_{2J,0} - P_{0,2J} = \frac{zw\,[\chi_{2J-1}(z) + \chi_{2J-1}(w)]}{(z-w)(zw-1)}. \tag{18}$$

Proof. Insert $\chi_n(z) = (z^{n+1} - z^{-n-1})/(z - z^{-1})$ into (15) and sum the four geometric series, giving (16). Subtraction and cancellation give (17)–(18). By Lemma 2.3, these Abel values are the eta values at zero despite the different positive spectral shifts. □

For $m \geq 0$, let

$$W_m^X = \mathrm{Sym}^m(\mathbb{C}^2) \circ \tau_X, \qquad (A_m^X)_{\rho\sigma} = \dim \mathrm{Hom}_G(\rho, W_m^X \otimes \sigma), \quad X = L, R. \tag{19}$$

Both matrices act on the representation ring of the full deck group G; neither projection is required to be faithful.

Theorem 3.2 (General free-Spin(4) Dirac-to-TT transfer)**.** *Under the hypotheses of Section 2, for every integer $J \geq 1$,*

$$\epsilon_J(g) = -\frac{1}{2}\left[\chi_{2J-1}(\tau_L(g)) + \chi_{2J-1}(\tau_R(g))\right] d_C(g), \tag{20}$$

$$\boxed{\boldsymbol{\eta}_J^{\mathrm{TT}} = -\frac{1}{2}\left(A_{2J-1}^L + A_{2J-1}^R\right)\boldsymbol{\eta}_C^D.} \tag{21}$$

This is an identity of full eta invariants with copy-multiplicity normalization.

Equation (21) relates meromorphically continued eta invariants. The Dirac and TT spectra lie on different arithmetic progressions, so the formula is neither a shellwise relation nor an operator intertwining or isospectrality statement.

Proof. For a noncentral pair, (20) is the quotient of (18) by (17). Removable Weyl limits use

$$\chi_m(1) = m + 1, \qquad \chi_m(-1) = (-1)^m(m+1). \tag{22}$$

If exactly one coordinate is central and the other is not, the corresponding single Weyl limit exists; freeness keeps the simplified denominator nonzero, so (20) still holds. The two fully central mixed cases are

$$(-1,1):\ d_C = \frac{1}{2},\ \epsilon_J = 0, \qquad (1,-1):\ d_C = -\frac{1}{2},\ \epsilon_J = 0. \tag{23}$$

At $(1,1)$ the signed Dirac and TT shells vanish termwise. The same shellwise cancellation occurs at $(-1,-1)$, but that pair is excluded by effectivity.

For clarity, all fully central possibilities are:

(τ_L, τ_R)	d_C	ϵ_J	$-\frac{1}{2}(\chi^L_{2J-1} + \chi^R_{2J-1})$	status
$(1,1)$	0	0	$-2J$	identity
$(-1,-1)$	0	0	$+2J$	ineffective, excluded
$(-1,1)$	$\frac{1}{2}$	0	0	free mixed class allowed
$(1,-1)$	$-\frac{1}{2}$	0	0	free mixed class allowed

The irreducible characters are an orthonormal basis for class functions. Multiplication by $\chi(W_m^X)$ has matrix A_m^X in that basis, so projection of (20) using (6) gives (21); no factor $\dim \rho$ appears. □

Remark 3.3 (Spin-lift covariance)**.** Two lifts differ by a sign character $\lambda : G \to \{\pm 1\}$. Simultaneous central multiplication gives

$$d_C^{\lambda\tau} = \lambda d_C^{\tau}, \qquad W_m^{X,\lambda\tau} = \lambda^m \otimes W_m^{X,\tau} \quad (m \geq 0),$$

whereas every TT shell has even total central parity and is unchanged. Thus if P_λ denotes the permutation of irreducible coordinates induced by tensoring with λ, then

$$(\boldsymbol{\eta}_C^D)^{\lambda\tau} = P_\lambda (\boldsymbol{\eta}_C^D)^{\tau}, \qquad A_m^{X,\lambda\tau} = P_\lambda^m A_m^{X,\tau} = A_m^{X,\tau} P_\lambda^m, \qquad P_\lambda^2 = I.$$

Consequently, for general $m \geq 0$,

$$A_m^{X,\lambda\tau} (\boldsymbol{\eta}_C^D)^{\lambda\tau} = P_\lambda^{m+1} A_m^{X,\tau} (\boldsymbol{\eta}_C^D)^{\tau}.$$

For the odd powers $m = 2J-1$ used in (21), the factor $P_\lambda^{m+1} = P_\lambda^{2J} = I$ cancels. This cancellation is not asserted for even m. In particular, $A_0^{X,\tau} = I$ for either lift; replacing P_λ^m by P_λ would incorrectly give $I = P_\lambda$ for a nontrivial sign character. The integer-rank TT vector is lift-independent.

3.1 One-sided affine-ADE specialization

Now let $G = \Gamma < SU(2)$ act from the left, so $\tau_L(g) = g$ and $\tau_R(g) = 1$. Put $M_\Gamma = \Gamma\backslash S^3$ and let $\Delta = V_1$ be the defining representation. The general eta definition becomes

$$\eta_{J,\rho}^{\mathrm{TT}}(s) = \sum_{n \geq 0} \frac{(n+1)m_\rho(V_{n+2J}) - (n+2J+1)m_\rho(V_n)}{(n+J+1)^s}. \tag{24}$$

For a noncentral element with eigenvalues u, u^{-1}, set

$$F_J(u) = u\left[\frac{u(u^{2J}-1)}{(1-u)^2} + \frac{u^{2J} - 2J - 1}{1-u}\right]. \tag{25}$$

Then

$$\epsilon_J(g) = \frac{F_J(u) - F_J(u^{-1})}{u - u^{-1}}. \tag{26}$$

Character projection gives

$$\eta_{J,\rho}^{\mathrm{TT}}(M_\Gamma) = \frac{1}{|\Gamma|} \sum_{g \in \Gamma} \overline{\chi_\rho(g)} \epsilon_J(g). \tag{27}$$

The identity and central -1 classes have value zero by signed shellwise cancellation.

Define

$$(A_m)_{\rho\sigma} = \dim \mathrm{Hom}_\Gamma(\rho, V_m \otimes \sigma). \tag{28}$$

Thus $A_0 = I$, A_1 is the ordinary McKay matrix, and $A_{m+1} = A_1 A_m - A_{m-1}$. Since $A_{2J-1}^R = 2JI$, the general theorem gives the following corollary.

Corollary 3.4 (One-sided Dirac-to-TT transfer)**.** *For every finite* $\Gamma < SU(2)$ *and every integer* $J \geq 1$*,*

$$\boxed{\boldsymbol{\eta}_J^{\mathrm{TT}} = -\left(JI + \frac{1}{2}A_{2J-1}\right)\boldsymbol{\eta}_C^D.} \tag{29}$$

Equivalently, with

$$d_C(1) = 0, \qquad d_C(g) = \frac{2}{2 - \chi_\Delta(g)} \quad (g \neq 1), \tag{30}$$

one has $\epsilon_J(g) = -(J + \chi_{2J-1}(g)/2)d_C(g)$*, including the central values.*

There is an equivalent finite-representation form that makes the identity subtraction explicit. Define

$$P_J = \bigoplus_{r=0}^{2J-2} (2J - 1 - r)\,\mathrm{Sym}^r \Delta, \qquad \dim P_J = \frac{J(4J^2-1)}{3}. \tag{31}$$

Corollary 3.5 (Triangular representation form)**.** *For every* $\rho \in \mathrm{Irr}(\Gamma)$*,*

$$\boxed{\eta_{J,\rho}^{\mathrm{TT}} = -2J\eta_{C,\rho}^D + \langle \rho, P_J|_\Gamma \rangle - \frac{\dim \rho}{|\Gamma|}\frac{J(4J^2-1)}{3}.} \tag{32}$$

Proof. The $SU(2)$ character recurrence gives

$$(2 - x)\chi_{P_J} = 2J - \chi_{2J-1}. \tag{33}$$

Indeed, for $a_r = 2J - 1 - r$, the coefficient of χ_0 in $(2 - x)\sum_r a_r\chi_r$ is $2a_0 - a_1 = 2J$, every interior coefficient is $2a_r - a_{r-1} - a_{r+1} = 0$, and the terminal coefficient is -1. Combining (33) with (30), projecting, and removing the identity contribution gives (32). □

Remark 3.6 (Sign conventions)**.** Some references use the sign-reversed Dirac operator $-D_C$, which reverses the leading sign in the transfer, and reversing the two TT helicity branches negates $\boldsymbol{\eta}_J^{\mathrm{TT}}$. Every formula here uses the Cisneros–Molina Dirac sign and the TT branch order fixed in (4).

4 Lens-space specialization

A general lens space has a mixed cyclic action and is not another affine-A rung in the one-sided McKay classification. It is, however, an immediate cyclic specialization of Theorem 3.2. The resulting closed-form evaluations and detailed convention comparisons are developed in [Mah26b]; here we record the transfer itself.

Write

$$L(p,q) = S^3/\langle\gamma\rangle, \qquad \gamma(z_1,z_2) = (\zeta z_1, \zeta^q z_2), \qquad \zeta = \mathrm{e}^{2\pi i/p}, \quad \gcd(p,q) = 1.$$

Choose a lift bit ℓ satisfying

$$q + 1 + \ell p \equiv 0 \pmod 2 \tag{34}$$

and put $\widetilde{q} = q + \ell p$. The lifted left and right angles are

$$\alpha = \frac{\pi(1+\widetilde{q})}{p}, \qquad \beta = \frac{\pi(1-\widetilde{q})}{p}. \tag{35}$$

Thus the two factors of a lifted deck class are represented by $g_L = \mathrm{e}^{i\alpha}$ and $g_R = \mathrm{e}^{i\beta}$ in their defining tori. Odd p has one valid lift; even p has the two lifts $\widetilde{q} = q$ and $q + p$.

For this section the Dirac branches are

$$E_n^{D,+} = V_n \boxtimes V_{n+1}, \qquad E_n^{D,-} = V_{n+1} \boxtimes V_n,$$

and the TT branches are exactly (4). Let $d_C(g_L, g_R)$ denote the full Cisneros-sign equivariant Dirac eta density for these branches.

Corollary 4.1 (General lens-space transfer)**.** *For every $p \geq 2$, $\gcd(p, q) = 1$, every valid spin lift, and every $J \geq 1$, a nonidentity deck class obeys*

$$\boxed{\epsilon_J(g_L, g_R) = -\frac{1}{2}\left[\chi_{2J-1}(g_L) + \chi_{2J-1}(g_R)\right] d_C(g_L, g_R).} \tag{36}$$

After projection to all flat characters of $\mathbb{Z}_p$,

$$\boxed{\boldsymbol{\eta}_J^{\mathrm{TT}}(L(p,q)) = -\frac{1}{2}\left(A_{2J-1}^L + A_{2J-1}^R\right)\boldsymbol{\eta}_C^D(L(p,q)).} \tag{37}$$

The two matrices are the finite weight-convolution actions induced by the left and right lifted circle homomorphisms.

Proof. Substitute the lifted torus elements into (20). Their characters are

$$\chi_{2J-1}(g_L) = \frac{\sin(2J\alpha)}{\sin\alpha}, \qquad \chi_{2J-1}(g_R) = \frac{\sin(2J\beta)}{\sin\beta},$$

with the Weyl limits (22) at central factors. Fourier projection over $\mathbb{Z}_p$ gives (37). □

The finite convolution can be written explicitly. Set

$$s = \frac{1+\widetilde{q}}{2}, \qquad t = \frac{1-\widetilde{q}}{2} \pmod p,$$

and let j run through the odd weights $-(2J-1), -(2J-3), \ldots, 2J-1$. Then

$$\eta_{J,a}^{\mathrm{TT}} = -\frac{1}{2}\sum_j \left(\eta_{C,a-js}^D + \eta_{C,a-jt}^D\right). \tag{38}$$

For $q = 1$ and $\ell = 0$, one has $t = 0$, so $A_{2J-1}^R = 2JI$ and the one-sided theorem (29) is recovered exactly. Changing the even-order spin lift shifts the Dirac vector and both odd kernels by $p/2$; these shifts cancel in (38), so the integer-rank TT vector is lift-independent.

5 A mixed nonabelian example

A mixed nonabelian action is

$$G = Q_8 \times \mathbb{Z}_3, \qquad (a, u^r) \cdot x = a\, x\, u^{-r}. \tag{39}$$

The defining trace sets of the two projections are respectively $\{2, -2, 0\}$ and $\{2, -1\}$. Their sole common trace is 2, attained only by the identity in each factor, so the action is free; $(-1, -1)$ is absent, so it is effective. The group has 15 complex irreducible representations and four spin lifts. Write A_0, A_1, A_2, A_3, D for the four one-dimensional and one two-dimensional Q_8 irreps, and c_0, c_1, c_2 for the $\mathbb{Z}_3$ characters. In the manuscript convention, the rank-one values are

Q_8 irrep	$\eta_C^D(\rho \otimes c_0)$	$\eta_C^D(\rho \otimes c_1)$	$\eta_1^{\mathrm{TT}}(\rho \otimes c_0)$	$\eta_1^{\mathrm{TT}}(\rho \otimes c_1)$
A_0	$-89/144$	$103/144$	$-17/36$	$-5/36$
A_1, A_2, A_3	$55/144$	$-41/144$	$19/36$	$-5/36$
D	$-35/72$	$13/72$	$-4/9$	$2/9$

The c_2 column equals the c_1 column. These values illustrate the theorem; they do not prove it.

Example 5.1 (A non-self-dual coordinate and the matrix side)**.** The characters c_1 and $c_2 = c_1^*$ are genuinely complex and non-self-dual. The lifted defining representations are

$$W_1^L = D \otimes c_0, \qquad W_1^R = A_0 \otimes (c_1 \oplus c_2).$$

With the row index denoting the target copy and the column index the source copy in (19), the $A_0 \otimes c_1$ component of the rank-one transfer is therefore

$$\eta_1^{\mathrm{TT}}(A_0 c_1) = -\frac{1}{2}\left[\eta_C^D(A_0 c_0) + \eta_C^D(A_0 c_2) + \eta_C^D(D c_1)\right].$$

The table gives

$$-\frac{1}{2}\left[-\frac{89}{144} + \frac{103}{144} + \frac{13}{72}\right] = -\frac{5}{36},$$

exactly the displayed TT coordinate. This calculation fixes the associated-bundle convention, shows why the multiplication matrix acts without a transpose, and exhibits directly that no factor $\dim(A_0 c_1)$ or $\dim D$ is inserted. Relabeling every coordinate by its dual exchanges c_1 and c_2 and leaves the vector theorem unchanged.

6 The affine-ADE source recurrence

Let $N = A_1$ be the McKay matrix and put $E_J = (\eta_{J,\rho}^{\mathrm{TT}})_{\rho \in \mathrm{Irr}(\Gamma)}$. For a noncentral class with defining eigenvalues u, u^{-1}, define the recurrence residual

$$r_J(g) = \epsilon_{J+1}(g) + \epsilon_{J-1}(g) - \chi_\Delta(g)\epsilon_J(g), \qquad J \geq 2, \tag{40}$$

and put $r_J(\pm 1) = 0$. Substitution of (26) and the $SU(2)$ character recurrence gives

$$r_J(u) = -2J + \frac{(u^2+u+1)(u^{2J} - u^{-2J})}{u^2 - 1}, \qquad u \neq \pm 1, \tag{41}$$

and set

$$R_{J,\rho} = \frac{1}{|\Gamma|} \sum_{g \in \Gamma} \overline{\chi_\rho(g)} r_J(g). \tag{42}$$

Theorem 6.1 (Sourced McKay recurrence)**.** *For $J \geq 2$,*

$$\boxed{E_{J+1} + E_{J-1} - N E_J = R_J.} \tag{43}$$

Moreover,

$$\sum_\rho \dim(\rho) E_{J,\rho} = 0, \qquad \sum_\rho \dim(\rho) R_{J,\rho} = 0. \tag{44}$$

For every finite Γ, there is a vector v and a periodic vector b_J such that

$$R_J = Jv + b_J. \tag{45}$$

Proof. Project the defining identity (40) against the irreducible characters. Multiplication by χ_Δ becomes N, giving (43); direct algebra in u is the displayed (41). The regular character vanishes off the identity, while ϵ_J and r_J vanish at the identity, proving (44). Every noncentral eigenvalue u is a root of unity. The $u^{\pm 2J}$ part is therefore periodic in J, while the explicit first term is linear. Finite character transformation gives (45). □

The shape "linear plus periodic" has a classical antecedent at the multiplicity level for the spin-Dirac problem [CM01]. The claim here is the eta-vector lift for the integer-rank projected-TT family, not novelty for the underlying Clebsch–Gordan or McKay recurrence.

7 Affine A: cyclic groups

For $\Gamma = \mathbb{Z}_p$ in the one-sided action, the irreducible representations are the flat characters χ_c, $c \in \mathbb{Z}/p\mathbb{Z}$, and the McKay graph is $\widetilde{A}_{p-1}$. Put $d = \min(c, p-c)$. In the nonoverlap chamber $p \geq 4J-3$ for $J \geq 2$ (and $p \geq 2$ for $J = 1$), the exact twisted formula is

$$\boxed{\eta^{\mathrm{TT}}_{J,c}(L(p,1)) = -\frac{J}{3p}\left(p^2 - 6pd + 6d^2 + 4J^2 - 2\right) + \left\lfloor \frac{\max(2J-d,0)^2}{4} \right\rfloor}. \tag{46}$$

Outside this chamber the exact finite character formula (27), or equivalently the transfer theorem, remains valid. The first excluded row $p = 4J-4$ differs only at the self-conjugate character $c = p/2$, by $+1$.

8 Affine D: every binary-dihedral group

Let

$$\mathrm{Dic}_m = \langle a, x \mid a^{2m} = 1,\ x^2 = a^m,\ xax^{-1} = a^{-1} \rangle, \qquad m \geq 2. \tag{47}$$

Its McKay graph is $\widetilde{D}_{m+2}$. Write

$$h^{\mathrm{cyc}}_{J,c} = \eta^{\mathrm{TT}}_{J,c}(L(2m,1)), \qquad c \in \mathbb{Z}/(2m),$$

for the cyclic value in the same orientation, generator, associated-bundle, and full-eta convention. When the value is represented through the Dirac transfer, we use the canonical one-sided spin lift.

The four one-dimensional representations are

$$A_t(a) = 1, \quad A_t(x) = t, \qquad t = \pm 1,$$

and

$$B_t(a) = -1, \quad B_t(x) = t, \qquad t = \pm i^m.$$

Thus $t^2 = (-1)^m$, as required by $x^2 = a^m$. The remaining irreducibles are two-dimensional:

$$V_r(a) = \begin{pmatrix} \zeta^r & 0 \\ 0 & \zeta^{-r} \end{pmatrix}, \qquad V_r(x) = \begin{pmatrix} 0 & (-1)^r \\ 1 & 0 \end{pmatrix}, \quad \zeta = \mathrm{e}^{\pi i/m}, \quad 1 \leq r < m.$$

Theorem 8.1 (Binary-dihedral all-irrep corollary)**.** *For every $m \geq 2$ and $J \geq 1$,*

$$\eta^{\mathrm{TT}}_{J,A_t} = \frac{1}{2} h^{\mathrm{cyc}}_{J,0} - \frac{tJ}{2}, \tag{48}$$

$$\eta^{\mathrm{TT}}_{J,B_t} = \frac{1}{2} h^{\mathrm{cyc}}_{J,m}, \tag{49}$$

$$\eta^{\mathrm{TT}}_{J,V_r} = h^{\mathrm{cyc}}_{J,r}. \tag{50}$$

These formulas remain valid without a stable-chamber restriction.

Proof. The cyclic subgroup $\langle a \rangle$ has order $2m$. The restrictions are $A_t|_{\langle a \rangle} = \chi_0$, $B_t|_{\langle a \rangle} = \chi_m$, and $V_r|_{\langle a \rangle} = \chi_r \oplus \chi_{-r}$. Every element of the coset $x\langle a \rangle$ has defining eigenvalues $+i, -i$, where the TT density is $-J$. Its character contribution is $-tJ/2$ on A_t, cancels on B_t, and vanishes on V_r. Finally $h^{\mathrm{cyc}}_{J,r} = h^{\mathrm{cyc}}_{J,2m-r}$, giving (48)–(50). □

At $m = 2$, $\mathrm{Dic}_2 = Q_8$ and the graph is $\widetilde{D}_4$. The two-dimensional defining-irrep eta vanishes for every rank; the four line representations carry the nontrivial source. For general m, the source and rank shifts are explicit. Put

$$Q_{J,c} = h^{\mathrm{cyc}}_{J+1,c} + h^{\mathrm{cyc}}_{J-1,c} - h^{\mathrm{cyc}}_{J,c-1} - h^{\mathrm{cyc}}_{J,c+1}, \qquad c \pmod{2m}.$$

Corollary 8.2 (Binary-dihedral source)**.** *Under the node labels of Theorem 8.1,*

$$R_{J,A_t} = \frac{1}{2} Q_{J,0} - tJ, \qquad R_{J,B_t} = \frac{1}{2} Q_{J,m}, \qquad R_{J,V_r} = Q_{J,r}. \tag{51}$$

The source after subtraction of its linear part has period dividing m.

Corollary 8.3 (Binary-dihedral rank shift)**.** *For the representatives $0 \leq c < 2m$,*

$$h^{\mathrm{cyc}}_{J+m,c} - h^{\mathrm{cyc}}_{J,c} = -\frac{4m^2 - 1}{12} + \frac{c(2m-c)}{2} + \frac{(-1)^c}{4}. \tag{52}$$

Consequently

$$\begin{aligned}
\eta^{\mathrm{TT}}_{J+m,A_t} - \eta^{\mathrm{TT}}_{J,A_t} &= -\frac{m^2 - 1}{6} - \frac{tm}{2}, \\
\eta^{\mathrm{TT}}_{J+m,B_t} - \eta^{\mathrm{TT}}_{J,B_t} &= \frac{2m^2 + 1 + 3(-1)^m}{24}, \\
\eta^{\mathrm{TT}}_{J+m,V_r} - \eta^{\mathrm{TT}}_{J,V_r} &= -\frac{4m^2 - 1}{12} + \frac{r(2m-r)}{2} + \frac{(-1)^r}{4}.
\end{aligned} \tag{53}$$

Proof. Write η^D_c for the cyclic Dirac eta value in the canonical one-sided lift. The transfer identity (29) gives

$$h^{\mathrm{cyc}}_{J+m,c} - h^{\mathrm{cyc}}_{J,c} = -m\eta^D_c - \sum_{\substack{j \bmod 2m \\ j \text{ odd}}} \eta^D_{c-j}.$$

Indeed, the $2m$ added weights in the symmetric-power character cover each odd residue modulo $2m$ twice. For $0 \leq c < 2m$, character projection of the one-sided Dirac density (30) evaluates to

$$\eta^D_c = \frac{4m^2 - 1}{12m} - \frac{c(2m-c)}{2m}.$$

Summing this quadratic over the odd residues gives $\sum_{j \text{ odd}} \eta^D_{c-j} = -(-1)^c/4$, where subscripts are reduced modulo $2m$. Substitution proves (52); Theorem 8.1 then gives the three binary-dihedral shifts. □

9 Affine E: the exceptional binary polyhedral groups

For the binary tetrahedral, octahedral, and icosahedral groups, the same class-transform and transfer theorems give the affine E_6, E_7, and E_8 eta vectors. Table 1 records the explicit atlas scope.

Group	Order	Irreps	Graph	Source structure
$2T$	24	7	$\widetilde{E}_6$	linear plus vector period 3
$2O$	48	8	$\widetilde{E}_7$	linear plus vector period 12
$2I$	120	9	$\widetilde{E}_8$	linear plus vector period 15

Table 1: Exceptional affine-E families and the periods of their eta-vector source terms.

For $2I$, the conjugacy-class sizes are

$$(1, 1, 30, 20, 20, 12, 12, 12, 12),$$

with defining traces

$$2, -2, 0, 1, -1, \varphi, \varphi^{-1}, -\varphi^{-1}, -\varphi.$$

In the irrep order

$$(\mathrm{triv}, \Delta, \Delta_2, 3\mathrm{D}, 3\mathrm{D}_2, 4\mathrm{S}, 4\mathrm{A}5, 5\mathrm{A}5, 6\mathrm{S}),$$

the dimension vector $d = (1, 2, 2, 3, 3, 4, 4, 5, 6)$ satisfies $Nd = 2d$. The rank source has vector period 15. Analogous class formulas and periods hold for $2T$ and $2O$. Complete rational tables are omitted for space. At spin $\frac{1}{2}$ such exceptional tables are classical [CM01]; the present extension is the projected-TT integer-rank eta tower.

10 Numerical verification

We checked the finite specializations with independent exact-arithmetic implementations. Direct spectral evaluation and the transfer formula agree for every flat twist in forty lifted lens-space cases, for all 120 coordinates of the $Q_8 \times \mathbb{Z}_3$ example at ranks $1, \ldots, 8$, and throughout the cyclic, binary-dihedral, and binary-polyhedral tables used here. These computations check the sign, lift, and copy-multiplicity conventions; the all-rank statements rest on the proofs above.

11 Antecedents, novelty perimeter, and exclusions

Flat-twisted eta invariants, equivariant fixed-point formulas, and the rank-one coexact curl sector are classical [APS75, Don78, Gil84, Bär19]. So are spin-Dirac eta on spherical space forms and its complete exceptional-group tables [Sea81, Bär92, Bär96, Goe97, CM01], the Molien-series approach [Deg09, Deg01], and the Clebsch–Gordan and McKay recurrences [McK80, Kos85, Ros03]. Higher-spin sphere spectra and quotient-character methods provide the geometric and representation-theoretic background [Hom01, HT21, Dow07, CLM11]; charged spin-$\frac{1}{2}$ and spin-$\frac{3}{2}$ lens-space formulas provide related half-integer antecedents [BL23, DDHM23].

The rank-one perimeter is especially strict. Millson's equivariant curl calculation is recorded here through Bär's full-text attribution, and Bär treats curl spectra for general spherical space forms [Bär19]. Capoferri–Vassiliev state the classical APS-derived relation, in their Dirac convention,

$$\eta_{\mathrm{curl}}(0) = -4\eta_{\mathrm{Dirac}}(0) + k, \qquad k \in \mathbb{Z},$$

on a general closed oriented three-manifold [CV26, Rem. D.4(ii)–(iii)]. Their Dirac operator carries the opposite sign to the operator whose positive branch is $V_n \boxtimes V_{n+1}$ in (11). In the present convention the same relation is $\eta_{\mathrm{curl}} \equiv +4\eta_C^D \pmod{\mathbb{Z}}$. For example, on $L(p,1)$ with the canonical one-sided lift $\tau_R = 1$, it reads $\eta_{\mathrm{curl}} - 4\eta_C^D = -(p-1)$. The transfer identity is not a determination of the integer in the general closed-manifold relation. Thus the rank-one, untwisted, modulo-integers shadow of (21) is not new. Fixed-rank constructions of higher-spin eta from twisted Dirac data in the Rarita–Schwinger anomaly literature are also mechanism antecedents; the claim here is the exact, mod-$\mathbb{Z}$-free, integer-rank vector identity with arbitrary flat irreducible twists.

Eta invariants at flat connections also enter gauge Chern–Simons theory classically [Jef92]; we do not claim a new connection to that theory.

Cisneros–Molina also gives a linear-plus-periodic multiplicity recurrence in the spectral label. Theorem 6.1 must therefore be framed as an eta-level lift to an operator family indexed by integer rank, not as the first appearance of linear-periodic representation data. Likewise, the new claims concerning the exceptional families are confined to the projected-TT higher-rank transfer and source recurrence at $J \geq 2$.

To the best of our knowledge, the all-rank identities (21) and (43) are new for $J \geq 2$, as is the genuinely mixed two-sided form of Theorem 3.2. This is not a claim of absolute priority.

The results are restricted to free, effective, orientation-preserving round quotients with a specified spin lift. They do not treat orbifolds, off-round metrics, or a complete determinant or numerical APS index. No physical interpretation is attached to these results.

12 Discussion and outlook

The transfer theorem changes the computational organization of the problem. Without it, each rank and twist appears to require a new infinite signed spectral continuation. With it, the analytic work is concentrated in the classical Dirac vector; every integer-rank TT eta vector follows by finite representation-ring algebra. The affine-ADE diagrams then organize the calculation rather than merely label the deck groups.

The theorem supplies exact compact-manifold phase data and finite recurrences that may enter a properly gauge-fixed determinant or topological field theory calculation. Constructing such a theory would additionally require ghosts, zero modes, spectral cuts, and local counterterms, and lies beyond the present spectral result.

The classwise identity now covers every free effective finite round action with a chosen Spin(4) lift. Three further directions are natural. First, one can turn the theorem into explicit irrep tables for the remaining mixed nonabelian spherical-space-form families; this is an atlas problem, not a new transfer theorem. Second, one can study metric deformation away from the round quotient. Third, the eta transfer may be combined with determinant magnitudes only after fixing ghosts, zero modes, spectral cuts, and local counterterms. Each requires separate analysis and none follows from the present theorem.

A companion result realizes $\mathcal{C}_J$ naturally as the TT reducing summand of the tangential operator of a four-dimensional chiral twisted-Dirac problem on round spherical quotients. That boundary realization is logically independent of the eta-transfer proof and is not used here. In particular, it does not identify $\eta(\mathcal{C}_J)$ with the whole APS boundary correction: the complementary boundary summand and its kernel remain separate terms [Mah26c]. The branchwise all-s recurrences, pole laws, heat expansions, and relative determinants are likewise developed in the separate bivariate McKay manuscript rather than duplicated here [Mah26a].

A Character identity behind the transfer

For completeness, the triangular identity (33) can be checked without spectral analysis. Write

$$\chi_{P_J} = \sum_{r=0}^{2J-2} (2J-1-r)\chi_r$$

and use $x\chi_r = \chi_{r+1} + \chi_{r-1}$ with $\chi_{-1} = 0$. Every interior coefficient cancels. The two endpoints leave $2J\chi_0$ and $-\chi_{2J-1}$. This proof also shows that P_J is finite and that

$$\dim P_J = \sum_{r=0}^{2J-2} (2J-1-r)(r+1) = \frac{J(4J^2-1)}{3}.$$

B Regular completeness

Let $\mathcal{R} = \bigoplus_\rho (\dim \rho)\rho$ be the regular representation. Summing an eta vector with weights $\dim \rho$ is the same as taking the regular character. It vanishes off the identity; the signed sphere eta density vanishes at the identity. Hence

$$\sum_\rho \dim \rho \, \eta^{\mathrm{TT}}_{J,\rho} = 0.$$

The identical argument applies to the source vector. This is the correct completeness normalization; replacing copy multiplicity by isotypic dimension would generally spoil it.

Acknowledgment of assistance

AI-assisted research, computation, checking, and editorial support used OpenAI Codex and Anthropic Claude. Luqman Mahmood directed the work and is responsible for the claims and final text. These internal checks are not external peer review.

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